Six Birds: No-Go Theorems for Audited Emergence: Supplement

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How to use this supplement

This supplement collects the repo-facing apparatus that accompanies the main theorem paper. It has two purposes. First, it records provenance, evidence bindings, reproducibility commands, and fragility guardrails in one place. Second, it summarizes the distribution of Lean support across the eight theorem fronts without interrupting the mathematical flow of the main text.

The main paper is intended to stand on its own as a theorem paper. The present supplement is therefore organized around audit and provenance questions: where each claim is bound, which frozen assets support it, which reproduce steps build the relevant summaries, and where the formal boundary is explicitly frozen.

1 Provenance narrative

The theorem line is assembled from a fixed project spine. The theorem order and section homes come from `docs/project/theorem_atlas.yaml`. The formal statements used in the LaTeX theorem blocks come from `docs/project/theorem_statement_sheet.yaml`, and notation is stabilized in `docs/project/notation_registry.yaml`.

Claim-level bindings are frozen in `docs/project/claim_audit_freeze.yaml` and `docs/project/theorem_evidence_map.yaml`. Asset choices are frozen in `docs/project/paper_asset_freeze.yaml` and `docs/project/caption_registry.yaml`. Readiness and formal-support status are tracked in `docs/project/readiness_checklist.yaml` together with the per-front Lean bridge files under `docs/project/lean_*.yaml`.

The supplement tables below therefore do not introduce new mathematical claims. They record how the frozen theorem fronts, claim rows, assets, and executable audit artifacts are aligned in the packaged repo state.

2 Theorem–evidence map

This section records the frozen claim bindings, evidence handles, and asset/callout decisions used by the theorem fronts. The first table is generated from `docs/project/theorem_evidence_map.yaml`, `docs/project/paper_asset_freeze.yaml`, and `docs/project/caption_registry.yaml`. The second table summarizes the frozen callout bindings and their primary sources.

Table 1: Claim-level theorem–evidence bindings.

Theorem	Claim	Primary handle	Evidence artifacts	Asset / callout decision
NG_ARROW_DPI	NG_ARROW_DPI.core	results/T08/arrow_metrics.json	results/T08/arrow_metrics.json; results/T15/initial_distribution_sweep.csv	FIG-2:repo_asset; Figure 2 ([FIG-2]). Hidden-clock DPI / protocol matrix: honest observ...
NG_PROTOCOL_TRAP	NG_PROTOCOL_TRAP.core	results/T15/initial_distribution_sweep.csv	results/T15/initial_distribution_sweep.csv; results/T08/arrow_metrics.json	FIG-2:repo_asset; Figure 2 ([FIG-2]). Hidden-clock DPI / protocol matrix: honest observ...
NG_FORCE_FOREST	NG_FORCE_FOREST.core	results/T14/primitive_coverage.json	results/T14/primitive_coverage.json; results/master/witness_metrics.csv	FIG-3:repo_asset; Figure 3 ([FIG-3]). Graph / affinity contrasts: forest support remove...
NG_FORCE_NULL	NG_FORCE_NULL.core	results/master/witness_metrics.csv	results/master/witness_metrics.csv	FIG-3:repo_asset; Figure 3 ([FIG-3]). Graph / affinity contrasts: forest support remove...
NG_MACRO_CLOSURE_DEFICIT	NG_MACRO_CLOSURE_DEFICIT.core	results/T09/closure_metrics.json	results/T09/closure_metrics.json; results/T09/grid_check.json	FIG-4:repo_asset; Figure 4 ([FIG-4]). Closure front: exact packaged future laws define ...
NG_OBJECT_CONTRACTIVE	NG_OBJECT_CONTRACTIVE.core	results/T10/objecthood_metrics.json	results/T10/objecthood_metrics.json; results/T10/stability_grid.json	FIG-5:repo_asset; Figure 5 ([FIG-5]). Objecthood front: in the admissible total-variati...
NG_LADDER_IDEM	NG_LADDER_IDEM.core	results/T11/no_ladder_demos.json	results/T11/no_ladder_demos.json; results/T11/definability_metrics.json	FIG-6:repo_asset; Figure 6 ([FIG-6]). Ladder front: fixed idempotent packaging saturate...
NG_LADDER_BOUNDED_INTERFACE	NG_LADDER_BOUNDED_INTERFACE.core	results/T11/definability_metrics.json	results/T11/definability_metrics.json; results/T11/no_ladder_demos.json	TBL-6:repo_asset; Table 6 ([TBL-6]). Ladder theorem evidence, count core, and corollary...
NG_ARROW_DPI	NG_ARROW_DPI.guardrail	results/T15/adversarial_cases.json	results/T15/adversarial_cases.json; results/T15/initial_distribution_sweep.csv	TBL-2:repo_asset; Table 2 ([TBL-2]). Arrow / protocol theorem evidence and guardrails: ...
NG_PROTOCOL_TRAP	NG_PROTOCOL_TRAP.guardrail	results/T15/initial_distribution_sweep.csv	results/T15/initial_distribution_sweep.csv	TBL-2:repo_asset; Table 2 ([TBL-2]). Arrow / protocol theorem evidence and guardrails: ...

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Theorem	Claim	Primary handle	Evidence artifacts	Asset / callout decision
NG_FORCE_FOREST	NG_FORCE_FOREST.guardrail	results/T15/threshold_sweep.csv	results/T15/threshold_sweep.csv; results/T14/primitive_coverage.json	TBL-3:repo_asset; Table 3 ([TBL-3]). Graph / affinity theorem evidence and guardrails: ...
NG_FORCE_NULL	NG_FORCE_NULL.guardrail	results/T15/threshold_sweep.csv	results/T15/threshold_sweep.csv; results/T14/primitive_coverage.json	TBL-3:repo_asset; Table 3 ([TBL-3]). Graph / affinity theorem evidence and guardrails: ...
NG_MACRO_CLOSURE_DEFICIT	NG_MACRO_CLOSURE_DEFICIT.guardrail	results/T09/grid_check.json	results/T09/grid_check.json; results/T09/closure_metrics.json	TBL-4:repo_asset; Table 4 ([TBL-4]). Closure theorem evidence and provenance: the direc...
NG_OBJECT_CONTRACTIVE	NG_OBJECT_CONTRACTIVE.guardrail	results/T15/objecthood_metric_sweep.csv	results/T15/objecthood_metric_sweep.csv; results/T10/stability_grid.json	TBL-5:repo_asset; Table 5 ([TBL-5]). Objecthood admissible regime and guardrail matrix:...
NG_LADDER_BOUNDED_INTERFACE	NG_LADDER_BOUNDED_INTERFACE.guardrail	results/T11/no_ladder_demos.json	results/T11/no_ladder_demos.json; results/T11/definability_metrics.json	FIG-6:repo_asset; Figure 6 ([FIG-6]). Ladder front: fixed idempotent packaging saturate...

Table 2: Frozen callout bindings and caption decisions.

Callout	Theorem ids	Primary source	Supporting refs	Caption / note
FIG-1	none	authored:theory-package-and-audit-convention-schematic	docs/project/theorem_atlas.yaml; docs/project/claim_audit_freeze.yaml	Figure 1 ([FIG-1]). Finite audited theory package and audit-convention schematic: microstate s...
FIG-2	NG_ARROW_DPI; NG_PROTOCOL_TRAP	results/T15/horizon_lens_sweep.csv	results/T08/arrow_metrics.json; results/T15/initial_distribution_sweep.csv	Figure 2 ([FIG-2]). Hidden-clock DPI / protocol matrix: honest observed path-reversal KL can s...
FIG-3	NG_FORCE_FOREST; NG_FORCE_NULL	results/T14/primitive_coverage.json	results/master/witness_metrics.csv	Figure 3 ([FIG-3]). Graph / affinity contrasts: forest support removes cycle-space obstruction...
FIG-4	NG_MACRO_CLOSURE_DEFICIT	results/T09/closure_metrics.json	results/T09/grid_check.json	Figure 4 ([FIG-4]). Closure front: exact packaged future laws define the finite variational ob...
FIG-5	NG_OBJECT_CONTRACTIVE	results/T10/objecthood_metrics.json	results/T10/stability_grid.json	Figure 5 ([FIG-5]). Objecthood front: in the admissible total-variation regime, contraction yi...
FIG-6	NG_LADDER_IDEM; NG_LADDER_BOUNDED_INTERFACE	results/T11/no_ladder_demos.json	results/T11/definability_metrics.json	Figure 6 ([FIG-6]). Ladder front: fixed idempotent packaging saturates after one step, bounded...

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Callout	Theorem ids	Primary source	Supporting refs	Caption / note
TBL-1	NG_ARROW_DPI; NG_PROTOCOL_TRAP; NG_FORCE_FOREST; NG_FORCE_NULL; +4 more	results/master/witness_metrics.csv	results/master/witness_manifest.json	Table 1 ([TBL-1]). Theorem front map and support snapshot: the eight theorem fronts, their sec...
TBL-2	NG_ARROW_DPI; NG_PROTOCOL_TRAP	results/T15/adversarial_cases.json	results/T15/initial_distribution_sweep.csv	Table 2 ([TBL-2]). Arrow / protocol theorem evidence and guardrails: the paper binds the arrow...
TBL-3	NG_FORCE_FOREST; NG_FORCE_NULL	results/T15/threshold_sweep.csv	results/T14/primitive_coverage.json	Table 3 ([TBL-3]). Graph / affinity theorem evidence and guardrails: exact-support cycle infor...
TBL-4	NG_MACRO_CLOSURE_DEFICIT	results/T09/grid_check.json	results/T09/closure_metrics.json	Table 4 ([TBL-4]). Closure theorem evidence and provenance: the direct pack uses exact variati...
TBL-5	NG_OBJECT_CONTRACTIVE	results/T15/objecthood_metric_sweep.csv	results/T10/stability_grid.json	Table 5 ([TBL-5]). Objecthood admissible regime and guardrail matrix: exact contraction metric...
TBL-6	NG_LADDER_IDEM; NG_LADDER_BOUNDED_INTERFACE	results/T11/definability_metrics.json	results/T11/no_ladder_demos.json	Table 6 ([TBL-6]). Ladder theorem evidence, count core, and corollary map: no-ladder demos, ex...
TBL-A1	NG_ARROW_DPI; NG_PROTOCOL_TRAP; NG_FORCE_FOREST; NG_FORCE_NULL; +4 more	results/master/witness_manifest.json	results/master/witness_metrics.csv	Table A1 ([TBL-A1]). Artifact and evidence provenance map: each frozen core or guardrail claim...
TBL-A2	NG_ARROW_DPI; NG_PROTOCOL_TRAP; NG_FORCE_FOREST; NG_FORCE_NULL; +4 more	derived:lean-support-and-proof-dependency-map	docs/project/readiness_checklist.yaml; docs/project/theorem_atlas.yaml	Table A2 ([TBL-A2]). Lean support and proof dependency map: seven fronts are direct, objecthoo...

3 Computational audit and artifact manifest

The canonical command for rebuilding the computational side of the project is `make reproduce`. That command drives the reproducibility pipeline, refreshes theorem-pack summaries, Lean summaries, and rewrite-lane validations, and writes a manifest at `results/manifest/artifact_manifest.json`. The pipeline is deterministic up to the frozen repo contents and toolchain recorded in the manifest.

Table 3: Reproducibility manifest overview.

Field	Value
Canonical command	`make reproduce`
Manifest status	success
Generated at UTC	2026-03-22T07:25:15Z
Total pipeline steps	62
Hashed files	350
Failed steps	0

The table below lists the theorem-pack and Lean-check steps that are directly relevant to theorem evidence in the main paper, together with the summary JSON outputs they refresh.

Table 4: Theorem-pack and Lean-check steps relevant to theorem evidence.

Step	Status	Command	Summary outputs
run_t15_robustness_suite	passed	python3 scripts/run_t15_robustness_suite.py	results/T15/fragility_summary.json
run_t22_dpi_protocol_pack	passed	python3 scripts/run_t22_dpi_protocol_pack.py	results/T22/dpi_protocol_pack_summary.json
run_t23_graph_affinity_pack	passed	python3 scripts/run_t23_graph_affinity_pack.py	results/T23/graph_affinity_pack_summary.json
run_t24_closure_pack	passed	python3 scripts/run_t24_closure_pack.py	results/T24/closure_pack_summary.json
run_t25_objecthood_pack	passed	python3 scripts/run_t25_objecthood_pack.py	results/T25/objecthood_pack_summary.json
run_t26_bounded_interface_pack	passed	python3 scripts/run_t26_bounded_interface_pack.py	results/T26/bounded_interface_pack_summary.json
run_t27_lean_graph_bridge	passed	python3 scripts/run_t27_lean_graph_bridge.py	results/T27/lean_graph_bridge_summary.json
run_t28_lean_finite_lens_count	passed	python3 scripts/run_t28_lean_finite_lens_count.py	results/T28/lean_finite_lens_count_summary.json
run_t29_lean_bounded_interface_corollary	passed	python3 scripts/run_t29_lean_bounded_interface_corollary.py	results/T29/lean_bounded_interface_corollary_summary.json
run_t30_lean_objecthood_uniqueness	passed	python3 scripts/run_t30_lean_objecthood_uniqueness.py	results/T30/lean_objecthood_uniqueness_summary.json
run_t31_paper_skeleton	passed	python3 scripts/run_t31_paper_skeleton.py	results/T31/paper_skeleton_summary.json
run_t32_asset_freeze	passed	python3 scripts/run_t32_asset_freeze.py	results/T32/asset_freeze_summary.json
run_t33_draft_v1	passed	python3 scripts/run_t33_draft_v1.py	results/T33/draft_v1_summary.json
run_t34_red_team_review	passed	python3 scripts/run_t34_red_team_review.py	results/T34/red_team_summary.json
run_t35_scope_charter	passed	python3 scripts/run_t35_scope_charter.py	results/T35/lean_probability_scope_charter_summary.json
run_t36_lean_probability_core	passed	python3 scripts/run_t36_lean_probability_core.py	results/T36/lean_probability_core_summary.json
run_t37_lean_kl_dpi_core	passed	python3 scripts/run_t37_lean_kl_dpi_core.py	results/T37/lean_kl_dpi_core_summary.json
run_t38_lean_arrow_dpi	passed	python3 scripts/run_t38_lean_arrow_dpi.py	results/T38/lean_arrow_dpi_summary.json
run_t39_lean_protocol_trap	passed	python3 scripts/run_t39_lean_protocol_trap.py	results/T39/lean_protocol_trap_summary.json
run_t40_closure_feasibility	passed	python3 scripts/run_t40_closure_feasibility.py	results/T40/lean_closure_feasibility_summary.json
run_t41_lean_closure_variational_core	passed	python3 scripts/run_t41_lean_closure_variational_core.py	results/T41/lean_closure_variational_core_summary.json
run_t42_lean_closure_direct_pack	passed	python3 scripts/run_t42_lean_closure_direct_pack.py	results/T42/lean_closure_direct_pack_summary.json

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Step	Status	Command	Summary outputs
run_t43_objecthood_tv_feasibility	passed	python3 scripts/run_t43_objecthood_tv_feasibility.py	results/T43/lean_objecthood_tv_feasibility_summary.json
run_t44_lean_objecthood_direct_pack	passed	python3 scripts/run_t44_lean_objecthood_direct_pack.py	results/T44/lean_objecthood_direct_pack_summary.json

4 Fragility diagnostics

The main paper states theorem objects in exact finite terms. The fragility layer records where nearby proxy choices can distort interpretation without changing the underlying theorem statements. The frozen source for this section is `results/T15/fragility_summary.json`, together with the guardrail claims in `docs/project/claim_audit_freeze.yaml`.

The dominant risks are: changing the initial law in stationary arrow claims, changing the observation lens, fitting first-order macro proxies, thresholding exact support graphs, and switching to non-admissible objecthood metrics. Each such move is a scope change or a proxy diagnostic, not an in-theorem modification of the audited object.

Table 5: Fragility items and their frozen guardrails.

Fragility item	Severity	Affected claims	Guardrail claim	Mitigation / note
stationary_zero_arrow_is_fragile_to_initial_distribution	high	NG_PROTOCOL_TRAP.core	NG_PROTOCOL_TRAP.guardrail	Protocol-trap claims must be stated on the lifted chain or honest observed path law, under sta...
macro_arrow_depends_strongly_on_lens_choice	high	NG_ARROW_DPI.core; NG_PROTOCOL_TRAP.core	NG_ARROW_DPI.guardrail	Fitted first-order macro proxies are not theorem-level arrow audits and can create or distort ...
fitted_markov_proxy_can_create_or_distort_macro_arrow	high	NG_ARROW_DPI.guardrail; NG_PROTOCOL_TRAP.guardrail; NG_MACRO_CLOSURE_DEFICIT.guardrail	NG_ARROW_DPI.guardrail	Fitted first-order macro proxies are not theorem-level arrow audits and can create or distort ...
threshold_floors_can_introduce_spurious_one_way_support	medium	NG_FORCE_FOREST.guardrail; NG_FORCE_NULL.guardrail	NG_FORCE_NULL.guardrail	Thresholding or regularization can distort exactness and affinity interpretation; theorem clai...
object_multiplicity_can_be_metric_artifact	medium	NG_OBJECT_CONTRACTIVE.guardrail	NG_OBJECT_CONTRACTIVE.guardrail	Theorem objecthood claims must use admissible total-variation/Dobrushin metric assumptions; pr...

5 Formalization support map

This section summarizes the distribution of Lean support across the eight theorem fronts. The table is generated from `docs/project/readiness_checklist.yaml` together with the frozen Lean bridge files under `docs/project/lean_*.yaml`.

Table 6: Lean support distribution across theorem fronts.

Theorem	Lean status	Lean modules	Boundary note
NG_ARROW_DPI	direct	lean/SixBirdsNoGo/FiniteKLDPI.lean; lean/SixBirdsNoGo/FiniteKLDPIExample.lean; lean/SixBirdsNoGo/ArrowDPI.lean; +1 more	Direct support recorded in the frozen bridge files.
NG_PROTOCOL_TRAP	direct	lean/SixBirdsNoGo/ArrowDPI.lean; lean/SixBirdsNoGo/ArrowDPIExample.lean; lean/SixBirdsNoGo/ProtocolTrap.lean; +1 more	Direct support recorded in the frozen bridge files.
NG_FORCE_FOREST	direct	lean/SixBirdsNoGo/ForestForceBridge.lean; lean/SixBirdsNoGo/ForestForceBridgeExample.lean	Direct support recorded in the frozen bridge files.
NG_FORCE_NULL	direct	lean/SixBirdsNoGo/NullForceBridge.lean; lean/SixBirdsNoGo/NullForceBridgeExample.lean	Direct support recorded in the frozen bridge files.
NG_MACRO_CLOSURE_DEFICIT	direct	lean/SixBirdsNoGo/ClosureVariationalCore.lean; lean/SixBirdsNoGo/ClosureVariationalCoreExample.lean; lean/SixBirdsNoGo/ClosureDirectPack.lean; +1 more	Direct support recorded in the frozen bridge files.
NG_OBJECT_CONTRACTIVE	auxiliary_only	lean/SixBirdsNoGo/ContractionUniqueness.lean; lean/SixBirdsNoGo/ContractionUniquenessExample.lean	Boundary note: uniqueness core only; broader TV/Dobrushin plus epsilon-stability package is no...
NG_LADDER_IDEM	direct	lean/SixBirdsNoGo/Idempotence.lean	Direct support recorded in the frozen bridge files.
NG_LADDER_BOUNDED_INTERFACE	direct	lean/SixBirdsNoGo/Idempotence.lean; lean/SixBirdsNoGo/FiniteLensDefinability.lean; lean/SixBirdsNoGo/FiniteLensDefinabilityExample.lean; +2 more	Direct support recorded in the frozen bridge files.

For NG_OBJECT_CONTRACTIVE, the permanent boundary is explicit. The uniqueness-from-strict-contraction core is formalized, but the broader total-variation/Dobrushin plus ε-stability package used in the paper is not Lean-direct and remains frozen behind the objecthood TV boundary note.